

# Statistical empirical index of chemical weathering in igneous rocks: A new tool for evaluating the degree of weathering

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Received 17 October 2006; received in revised form 15 February 2007; accepted 19 February 2007

Editor: R.L. Rudnick

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## Abstract

Chemical weathering indices are useful tools in characterizing weathering profiles and determining the extent of weathering. However, the predictive performance of the conventional indices is critically dependent on the composition of the unweathered parent rock. To overcome this limitation, the present paper introduces an alternative statistical empirical index of chemical weathering that is extracted by the principal component analysis (PCA) of a large dataset derived from unweathered igneous rocks and their weathering profiles. The PCA analysis yields two principal components (PC1 and PC2), which capture 39.23% and 35.17% of total variability, respectively. The extent of weathering is reflected by variation along PC1, primarily due to the loss of Na<sub>2</sub>O and CaO during weathering. In contrast, PC2 is the direction along which the projections of unweathered felsic, intermediate and mafic igneous rocks appear to be best discriminated; therefore, PC1 and PC2 represent independent latent variables that correspond to the extent of weathering and the chemistry of the unweathered parent rock. Subsequently, PC1 and PC2 were then mapped onto a ternary diagram (MFW diagram). The *M* and *F* vertices characterize mafic and felsic rock source, respectively, while the *W* vertex identifies the degree of weathering of these sources, independent of the chemistry of the unweathered parent rock.

The *W* index has a number of significant properties that are not found in conventional weathering indices. First, the *W* index is sensitive to chemical changes that occur during weathering because it is based on eight major oxides, whereas most conventional indices are defined by between two and four oxides. Second, the *W* index provides robust results even for highly weathered sesquioxide-rich samples. Third, the *W* index is applicable to a wide range of felsic, intermediate and mafic igneous rock types. Finally, the MFW diagram is expected to facilitate provenance analysis of sedimentary rocks by identifying their weathering trends and thereby enabling a backward estimate of the composition of the unweathered source rock.

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**Keywords:** Chemical weathering; Weathering index; Logratio analysis; Principal component analysis

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## 1. Introduction

Weathering processes are one of the prime mechanisms that control the system of material recycling upon the Earth's surface, playing important roles in a variety

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of geological issues. These issues include (1) the regulation of long-term climate by the consumption of atmospheric CO<sub>2</sub> during the alteration of silicate materials (e.g., Berner and Kothavala, 2001), (2) the release of nutrients into the biosphere as a result of rock dissolution (e.g., Hamdan and Burnham, 1996), (3) the control of primary sediment yield derived from rock decomposition (e.g., McLennan, 1993), and (4) exploration for laterite resources such as bauxite and ferronickel ores (e.g., Gleeson et al., 2004). These fields of research are commonly investigated using geochemical indices that quantitatively reflect compositional changes attributed to weathering. Conventional weathering indices generally fall into two categories on the basis of their underlying approach, that is, they quantify the degree of weathering by either monitoring the decomposition of an unstable mineral or tracing the mass transfer of a labile element.

Examples of weathering indices that monitor the decomposition of an unstable mineral include the Chemical Index of Alteration (CIA; Nesbitt and Young, 1982), the Chemical Index of Weathering (CIW; Harnois, 1988) and the Plagioclase Index of Alteration (PIA; Fedo et al., 1995) (Table 1). Given that the CIA index logically predicts the extent of conversion of feldspar minerals and that feldspars are the most abundant rock-forming mineral group in the Earth's crust, this index has been widely employed in past studies (e.g., Taylor and McLennan, 1985; McLennan, 1993).

Other weathering indices evaluate the degree of chemical weathering by tracing the mass transfer of labile elements, as in the Weathering Index of Parker (WIP; Parker, 1970) or by making use simple molecular ratios such as, Al<sub>2</sub>O<sub>3</sub>/Fe<sub>2</sub>O<sub>3</sub>, SiO<sub>2</sub>/Fe<sub>2</sub>O<sub>3</sub> and SiO<sub>2</sub>/Al<sub>2</sub>O<sub>3</sub> (Jenny, 1941; Ruxton, 1968) (Table 1). Most of these indices exploit Al<sub>2</sub>O<sub>3</sub>, TiO<sub>2</sub> and Zr as normalizing elements on the assumption that their mass remained invariant during the weathering process. A clear advantage of such indices is that normalization enables the detection of changes in absolute elemental abun-

dance during weathering (e.g., Chadwick et al., 1990; Woronow and Love, 1990; White et al., 2001).

Despite their merits, the above geochemical indices are founded on restrictive assumptions that in many cases pose a formidable obstacle in practical use. The principal assumption involved in chemical indices that monitor the decomposition of an unstable mineral is that the initial geochemistry of the unweathered parent rock is invariant. For example, the CIA assumes Al<sub>2</sub>O<sub>3</sub>, Na<sub>2</sub>O, CaO and K<sub>2</sub>O reside exclusively in feldspars. If significant amounts of other minerals (e.g., amphiboles) that contain these elements were present in the rock, the application of the CIA becomes problematic. Consequently, different CIA values may simply reflect variations in the chemistry of the unweathered parent rock rather than the degree of weathering (e.g., Price and Velbel, 2003). Accordingly, the usefulness of these indices is strongly dependent on the geochemistry and petrology of the unweathered parent rock.

Indices that reflect the mass transfer of labile elements rely on the behavior of the specific element used in the ratio and do not consider the effects of other vital elements. During the course of weathering, elements generally behave in an inconsistent and even contradictory manner (e.g., Nesbitt and Young, 1984; Hamdan and Burnham, 1996; Nesbitt and Markovics, 1997; Price and Velbel, 2003) that is dependent upon the physico-chemical conditions (e.g., Hieronymus et al., 2001; White et al., 2001). For example, isovolumetric studies reveal that Fe is commonly leached during the early stages of weathering, but accumulates during the later stages (Hamdan and Burnham, 1996; Nesbitt and Markovics, 1997). Thus, the index value may change in an unexpected manner depending on the nature of the specific element used in the index. A reliable weathering index should ideally take into account of the behaviors of multiple elements to reduce the likelihood of obtaining biased results.

Recently, Duzgoren-Aydin et al. (2002), Duzgoren-Aydin and Aydin (2003) and Price and Velbel (2003) proposed that most of the chemical indices founded on

Table 1  
Representative chemical weathering indices

| Index | Formula <sup>a</sup>                                                                                                                          | Reference                |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| CIA   | $\text{Al}_2\text{O}_3 \times 100 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O} + \text{K}_2\text{O})$                        | Nesbitt and Young (1982) |
| PIA   | $(\text{Al}_2\text{O}_3 - \text{K}_2\text{O}) \times 100 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O} - \text{K}_2\text{O})$ | Fedo et al. (1995)       |
| CIW   | $\text{Al}_2\text{O}_3 \times 100 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O})$                                             | Harnois (1988)           |
| WIP   | $100[(2 \text{ Na}_2\text{O}/0.35) + (\text{MgO}/0.9) + (2 \text{ K}_2\text{O}/0.25) + (\text{CaO}^*/0.7)]$                                   | Parker (1970)            |
| SF    | $\text{SiO}_2/\text{Fe}_2\text{O}_3$                                                                                                          | Jenny (1941)             |
| R     | $\text{SiO}_2/\text{Al}_2\text{O}_3$                                                                                                          | Ruxton (1968)            |

<sup>a</sup> Molecular proportions of oxides. CaO\* denote CaO proportion derived from silicate minerals.

the assumptions described above are too tightly constrained to account for the complexities observed in actual weathering profiles. Given this limitation, it is desirable to establish a more flexible chemical index based on a completely different approach to that employed by existing indices. To our knowledge, few studies have attempted to determine a weathering index based on a statistical approach (e.g., Girty et al., 2003; von Eynatten et al., 2003). To this end, the present study proposes a new and alternative weathering index based on multivariate statistics; the index is established on a dataset derived from carefully investigated weathering profiles.

One desired property of a new weathering index is that it should be independent of the geochemical variability found in the unweathered parent rock types (Price and Velbel, 2003). It should also account for the behavior of multiple elements. These requirements can be satisfied with the use of principal component analysis, a statistical technique designed to extract orthogonal (independent) latent components expressed as a linear combination of multiple elements. We also introduce a ternary diagram delineated from the orthogonal coordinates of principal component analysis. This diagram depicts the weathering profiles of felsic to mafic igneous rocks upon a multivariate expansion of the conventional A-CN-K diagram which is generally used to display feldspar alteration and CIA values (Nesbitt and Young, 1984).

## 2. Materials and methods

### 2.1. Database for input

To uniquely define the latent variables that represent both geochemical variation within unweathered parent rock and variability induced by weathering, we collect data from the literature for unweathered felsic, intermediate and mafic igneous rocks as well as deeply weathered saprolites to laterites (Table 2). The samples obtained to represent unweathered rocks include 110 standard reference samples of felsic to mafic igneous rocks available from geological institutions such as the United States Geological Survey, Centre de Recherches Pétrographiques et Géochimiques, France and Geological Survey of Japan (Table 2). We assume that these standard materials represent the unweathered products of igneous rocks. In addition to the standard samples listed in Table 2, we also obtained the compositions of the common felsic to mafic igneous rocks provided by Le Maitre (1976), which gives 31 igneous rock average compositions that were calculated from a database of

more than 17,000 samples (these averages are probably arithmetic means).

Data on the 179 weathered samples considered in this study were obtained from analyses of recent saprolite and laterite profiles hosted within felsic to mafic rocks (Table 2). The present study restricts to analyze only the weathering profiles of igneous rocks and data from heterogeneous parent rocks (e.g., metamorphic rocks) are not considered. The weathering profiles listed in Table 2 represent weathering in the following climatic zones: tropical rainforest ( $N=18$ ), humid subtropical ( $N=41$ ), Mediterranean ( $N=41$ ), marine west coast ( $N=52$ ) and polar ice caps ( $N=27$ ). The database includes samples with a variety of weathering intensities, ranging from near-pristine to highly weathered.

The variables used in the principal component analysis were  $\text{SiO}_2$ ,  $\text{TiO}_2$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{Fe}_2\text{O}_3$ ,  $\text{MgO}$ ,  $\text{CaO}$ ,  $\text{Na}_2\text{O}$  and  $\text{K}_2\text{O}$  (wt.%). The major oxide components provided by different literature sources are not always consistent. This is especially true for  $\text{MnO}$ ,  $\text{P}_2\text{O}_5$ ,  $\text{FeO}$  and  $\text{LOI}$ . The  $\text{MnO}$  and  $\text{P}_2\text{O}_5$  concentrations in saprolite and laterite are commonly excluded because they fall below the detection limit, while some studies fail to report  $\text{LOI}$  content or distinguish between ferric and ferrous iron. Where  $\text{FeO}$  is reported in previous studies, we recalculated it as  $\text{Fe}_2\text{O}_3$  for entry into our database.  $\text{MnO}$ ,  $\text{P}_2\text{O}_5$  and  $\text{LOI}$  contents were omitted from our database and the remaining variables were re-closed to 100 wt.% in order to maintain consistency. For missing values and values of zero produced by rounding, we replaced these values with the maximum round-off errors (0.005%) according to the replacement operation advocated by Martín-Fernández et al. (2003). The replacement operation for rounding-derived values of zero is necessary because such values are not amenable to the following analyses.

Most weathering indices use the molecular proportions of oxides to calculate index values (e.g., Table 1). However, in the present analysis we used the weight percentages of oxides as the unit for input data. We took this approach because the statistical framework used in this study is scale-invariant and the results are identical irrespective of the chosen units (e.g., Aitchison, 1992). Accordingly, recalculation to molecular proportions is unnecessary because the two data formats return the same result.

### 2.2. Statistical framework

Principal component analysis (PCA) is a technique that combines numerous variables into several independent latent variables that underlie the multivariate data

Table 2  
Input data for principal component analysis

| Igneous standard reference samples <sup>a</sup> (N=110) |                               |                                    |                              |              |              |
|---------------------------------------------------------|-------------------------------|------------------------------------|------------------------------|--------------|--------------|
| Lithology                                               | Sample source and sample name |                                    |                              |              |              |
| Granite                                                 | ANRT GS-N                     | CRPG AC-E                          | CRPG GA                      | CRPG GH      | CRPG GR      |
|                                                         | GSJ JG-2                      | IAG YG-1                           | IGEM 2B                      | IGEM MK-3    | IGEM MK-4    |
|                                                         | IGGE GSR-1                    | IGI SG-1a                          | IRK GPMA401                  | IRK GPMA402  | IRK GPMA403  |
|                                                         | MINTEK NIM-G                  | MINTEK SARM48                      | RIAP DVG                     | RIAP SG-2    | RIAP SG-3    |
|                                                         | USGS G-1                      | USGS G-2                           | ZGI GM                       |              |              |
| Syenite                                                 | CCRMP SY-2                    | CCRMP SY-3                         | GSJ JSy-1                    | MINTEK NIM-S |              |
| Trachyte                                                | CRPG ISH-G                    | CRPG MDO-G                         |                              |              |              |
| Rhyolite                                                | GSJ JR-1                      | GSJ JR-2                           | USGS RGM-1                   |              |              |
| Rhyodacite                                              | RIAP DVR                      |                                    |                              |              |              |
| Granodiorite                                            | IGEM MK-1                     | IGEM MK-2                          | USGS GSP-1                   | USGS GSP-2   | GSJ JG-1     |
|                                                         | GSJ JG-1a                     | GSJ JG-3                           |                              |              |              |
| Plagiogranite                                           | IRK GPMA404                   |                                    |                              |              |              |
| Tonalite                                                | MRT T-1                       | USGS TLM-1                         |                              |              |              |
| Dacite                                                  | RIAP DVD                      |                                    |                              |              |              |
| Trachyrhyolite ( <i>sic</i> )                           | RIAP DVT                      |                                    |                              |              |              |
| Diorite                                                 | ANRT DR-N                     | IGI SKD-1                          |                              |              |              |
| Andesite                                                | GSJ JA-1                      | GSJ JA-2                           | GSJ JA-3                     | IGGE GSR-2   | USGS AGV-1   |
|                                                         | USGS AGV-2                    |                                    |                              |              |              |
| Hyalandesite                                            | RIAP DVA                      |                                    |                              |              |              |
| Basaltic andesite                                       | RIAP DVB                      |                                    |                              |              |              |
| Gabbro                                                  | BUL GV                        | CCRMP MRG-1                        | CCRMP WGB-1                  | CCRMP WMG-1  | GIT-IWG PM-S |
|                                                         | GO-ITA GOG-1                  | GSJ JGb-1                          | IGEM MO-3                    | IGEM MO-4    | IGEM MO-5    |
|                                                         | IGEM MO-8                     | IGEM MO-9                          | IGI SGD-1a                   | IGI SGD-2    | USGS GSM-1   |
| Diabase                                                 | CCRMP TDB-1                   | IGEM MO-1                          | IRK GPMA202                  | USGS W-1     | USGS W-2     |
| Basalt                                                  | CRPG BE-N CRPG BR             | IGEM MO-2                          | IGGE GSR-3                   | GSJ JB-1     |              |
|                                                         | GSJ JB-1a                     | GSJ JB-1b                          | GSJ JB-2                     | GSJ JB-3     | NIST SRM688  |
|                                                         | USGS BCR-1                    | USGS BCR-2                         | USGS BHVO-1                  | USGS BHVO-2  | USGS BIR-1   |
|                                                         | ZGI BM                        | NRCCRM GBW07126                    |                              |              |              |
| Dolerite                                                | GIT-IWG WS-E                  | IRK GPMA601                        | MINTEK SARM50                | USGS DNC-1   |              |
| Andesite-basalt                                         | IGEM MO-12                    |                                    |                              |              |              |
| Gabbro-diabase                                          | IRK GPMA201                   |                                    |                              |              |              |
| Gabbro-norite                                           | IRK GPMA203                   |                                    |                              |              |              |
| Metabasic rock                                          | WIHG MB-H                     |                                    |                              |              |              |
| Weathering profile (N=179)                              |                               |                                    |                              |              |              |
| Climatic zone <sup>b</sup>                              | Protolith                     | Locality                           | Reference                    |              |              |
| Tropical rainforest (Af)                                | Diorite                       | Luquillo Mountains, Puerto Rico    | Turner et al. (2003)         |              |              |
| Humid subtropical (Cfa)                                 | Granite                       | Fujian, China                      | Lan et al. (2003)            |              |              |
|                                                         | Granodiorite                  | Atlanta, USA                       | White et al. (2001)          |              |              |
| Mediterranean (Cs)                                      | Granodiorite                  | California, USA                    | Girty et al. (2003)          |              |              |
|                                                         | Granite                       | Serre, Italy                       | Mongelli (1993)              |              |              |
|                                                         | Andesite                      | Istanbul, Turkey                   | Ece and Nakagawa (2003)      |              |              |
| Marine west coast (Cfb)                                 | Granodiorite                  | Melbourne, Australia               | Nesbitt and Markovics (1997) |              |              |
|                                                         | Basalt                        | Victoria, Australia                | Price et al. (1991)          |              |              |
|                                                         | Basalt                        | Hamilton, New Zealand              | Moon and Jayawardane (2004)  |              |              |
|                                                         | Basalt                        | Massif Central, France             | Soubrand-Colin et al. (2005) |              |              |
| Polar ice cap (EF)                                      | Granodiorite                  | South Shetland Islands, Antarctica | Lee et al. (2004)            |              |              |
|                                                         | Andesite                      | South Shetland Islands, Antarctica | Lee et al. (2004)            |              |              |
|                                                         | Mafic lapilli tuff            | South Shetland Islands, Antarctica | Lee et al. (2004)            |              |              |

<sup>a</sup> Data from Flanagan (1986), Govindaraju (1994), Terashima et al. (1998), El-Maataoui et al. (1999) and Imai et al. (1999).

<sup>b</sup> Codes correspond to Köppen's climatic classification scheme (Köppen, 1923).

(as principal components). PCA can be viewed as a search for the orthogonal coordinates that explain the greatest amount of variation within the data. Since our

input data largely represent weathering profiles, it is expected that an extracted principal component will represent geochemical variation induced by weathering.

Ideally, this component should be independent of other principal components. If this is the case, the output will provide a full account of the behaviors of multiple elements and may enable the extraction of a latent variable that indicates the degree of weathering independent of the geochemistry of the unweathered parent rock. In undertaking the PCA and expressing the results graphically on a ternary diagram, we follow the approaches described by von Eynatten et al. (2003), Buccianti and Esposito (2004) and Tolosana-Delgado et al. (2005). These methods are based on the statistical analytical technique for compositional data described by Aitchison (1986) and Egozcue et al. (2003). In brief, the following sequence of operations was carried out (see Tolosana-Delgado et al., 2005: for more details, see the Appendix).

1. The geochemical compositional data, whose natural sample space is a simplex, were mapped onto a Euclidean real sample space using a logratio transformation (Aitchison, 1986). In the present case, the centered logratio transformation (clr; see Eq. (A2) in the Appendix) was adopted.
2. Following the clr transformation, the first and second principal components were extracted in the usual way via PCA. The PCA calculation was performed using singular value decomposition.
3. The PCA coordinates were back-transformed to the two-dimensional simplex sample space (ternary diagram) by inverse mapping of the isometric logratio transformation (ilr; see Eq. (A4) in the Appendix). Let  $v_1$  and  $v_2$  be orthogonal coordinates in the 2-dimensional real sample space, then the inverse transformation of ilr returns a three-part composition (Eq. (A4)). In this way, the orthogonal coordinates remain orthogonal within the framework of Aitchison geometry (Aitchison, 1992; Pawlowsky-Glahn and Egozcue, 2001).
4. Within the ternary diagram, a weathering trend can be described as a “compositional linear trend” (von Eynatten et al., 2003; von Eynatten, 2004). von Eynatten et al. (2003) demonstrated that weathering processes could be clearly explained from the compositional linear trends.

Despite the complexity of the above steps, the reason for their use is straightforward. The direct benefit following the logratio transformation is that compositional data are recast into a form that is suitable for standard PCA (e.g., Aitchison, 1986, 1990). Standard statistical concepts such as correlation, regression and PCA were originally developed for unconstrained data

within the Euclidean real sample space. As compositional data with a constant-sum constraint are a typical example of constrained data within the simplex sample space, the direct application of PCA to raw compositional data has the potential to produce inconsistent results (e.g., Davis, 2002, p. 523–526). An additional benefit is that logratio-transformed data commonly follow a multivariate normal distribution (Aitchison and Shen, 1980; Aitchison, 1986; Barceló-Vidal et al., 1996; Reymont, 1999; Weltje, 2002), which is another prerequisite of PCA (e.g., Atkinson et al., 2004). In contrast, raw compositional data, by definition, do not follow a multivariate normal distribution (Aitchison and Shen, 1980; Aitchison, 1986). These two points are the main reasons for adopting the statistical procedures outlined above, although this approach offers additional advantages in the treatment of compositional data (see Aitchison and Egozcue, 2005; Ohta and Arai, 2006). For example, the approach introduces proper classes of parametric distributions (e.g., additive logistic normal distribution; Aitchison and Shen, 1980) for the compositional data analysis and enables the simplex space to be reviewed as a metric space with a Euclidean structure and compositional data as an Abelian group (see Aitchison and Egozcue, 2005).

All statistical calculations were conducted on “R” (<http://www.R-project.org>) a free environment for statistical computing, using the package “compositions” (van den Boogaart and Tolosana-Delgado, 2006). Ternary plots were drawn using the software “ $\Delta$  plot”, which is freely distributed by Dr. Cedric M. John (<http://www.crog.org/cedric/dplot>).

### 3. Results

#### 3.1. PCA results

PCA loadings of clr-transformed data are presented in Table 3. The first principal component (PC1) shows a positive correlation with clr-transformed  $\text{SiO}_2$ ,  $\text{Al}_2\text{O}_3$  and  $\text{Fe}_2\text{O}_3$  components and negatively with  $\text{CaO}$  and  $\text{Na}_2\text{O}$  (Table 3). The second principal component (PC2) correlates positively with felsic oxides and negatively with mafic oxides. PC1 and PC2 captured 39.23% and 35.17% of the total variability, respectively. Collectively, 74.40% of the total variability was explained by these two components, with a much smaller amount being explained by the subsequent principal component (14.82% for PC3).

PC1 and PC2 are illustrated as a biplot (Gabriel, 1971; Aitchison and Greenacre, 2002) in Fig. 1. Samples collected from weathering profiles plot parallel



Table 3  
Results of principal component analysis

|                                | PC1    | PC2    | PC3    |
|--------------------------------|--------|--------|--------|
| SiO <sub>2</sub>               | 0.592  | 0.746  | 0.172  |
| TiO <sub>2</sub>               | 0.566  | −0.663 | 0.184  |
| Al <sub>2</sub> O <sub>3</sub> | 0.794  | 0.449  | 0.178  |
| Fe <sub>2</sub> O <sub>3</sub> | 0.618  | −0.584 | 0.440  |
| MgO                            | 0.146  | −0.528 | −0.805 |
| CaO                            | −0.820 | −0.292 | 0.426  |
| Na <sub>2</sub> O              | −0.857 | 0.430  | 0.134  |
| K <sub>2</sub> O               | 0.169  | 0.852  | −0.221 |
| Eigen value                    | 3.138  | 2.814  | 1.186  |
| Variance explained             | 39.23  | 35.17  | 14.82  |
| Cumulative variance            | 39.23  | 74.40  | 89.22  |

to the PC1 coordinate. A high positive PC1 score indicates intensively weathered samples, whereas unweathered igneous samples have zero to slightly negative PC1 scores. As an example, the “fresh granodiorite”, “weathered granodiorite” and “saprolite” samples of White et al. (2001) yield PC1 scores of −0.724 to −0.598, −0.619 to 2.369 and 2.548 to 3.877, respectively. The unweathered igneous samples plot perpendicular to the PC1, indicating that the PC1 is unaffected by chemical variability associated with igneous processes. Mafic and felsic rocks plot in the negative and positive fields of the PC2 coordinate, respectively (Fig. 1). In summary, the individual input data demonstrate that the PC1 correlates with the degree of weathering and the PC2 correlates with the unweathered parent rock types.

This interpretation was confirmed by examining the loadings of oxides. Fig. 1b shows the loadings as rays of oxides (solid blue arrows) and their links (dashed black arrows). Note that the interpretation of the biplot of clr-transformed compositional data is based on the links between variables, defined as vectors connecting the apices of the two corresponding rays (Aitchison and Greenacre, 2002). For example, if Al<sub>2</sub>O<sub>3</sub> is selected as a reference for the links, as in Fig. 1b, then the links are identical to forming the additive logratio (see Aitchison, 1986) with Al<sub>2</sub>O<sub>3</sub> as a common denominator. In this case, the links **Fe** and **Ti** are sub-parallel, and the links **Fe** and **Na** are sub-orthogonal. These relationships demonstrate that  $\ln(\text{Fe}_2\text{O}_3/\text{Al}_2\text{O}_3)$  and  $\ln(\text{TiO}_2/\text{Al}_2\text{O}_3)$  are strongly correlated and that  $\ln(\text{Fe}_2\text{O}_3/\text{Al}_2\text{O}_3)$  and  $\ln(\text{Na}_2\text{O}/\text{Al}_2\text{O}_3)$  show no correlation.

In Fig. 1b, the links **Na**, **Ca** and **K** are sub-parallel vectors with opposite directions to the PC1 coordinate. This relationship indicates that negative correlations exist between these links and PC1. Therefore, PC1 records the leaching of these soluble oxides (e.g., Na<sub>2</sub>O, CaO and K<sub>2</sub>O) relative to Al<sub>2</sub>O<sub>3</sub> proportion. In contrast,

the links **Fe** and **Ti** are approximately perpendicular to the PC1 coordinate. Hence, oxides such as Al<sub>2</sub>O<sub>3</sub>, Fe<sub>2</sub>O<sub>3</sub> and TiO<sub>2</sub> that commonly reside in weathered materials such as clay minerals and sesquioxide are invariant against PC1. All of these behaviors of elements in terms of PC1 are consistent with the weathering geochemistry of recent laterite and saprolite profiles (e.g., Nesbitt et al., 1980; Sharma and Rajamani, 2000; Hieronymus et al., 2001; White et al., 2001; Duzgoren-Aydin et al., 2002; Duzgoren-Aydin and Aydin, 2003), and thus, PC1 is interpreted to represent a weathering trend.

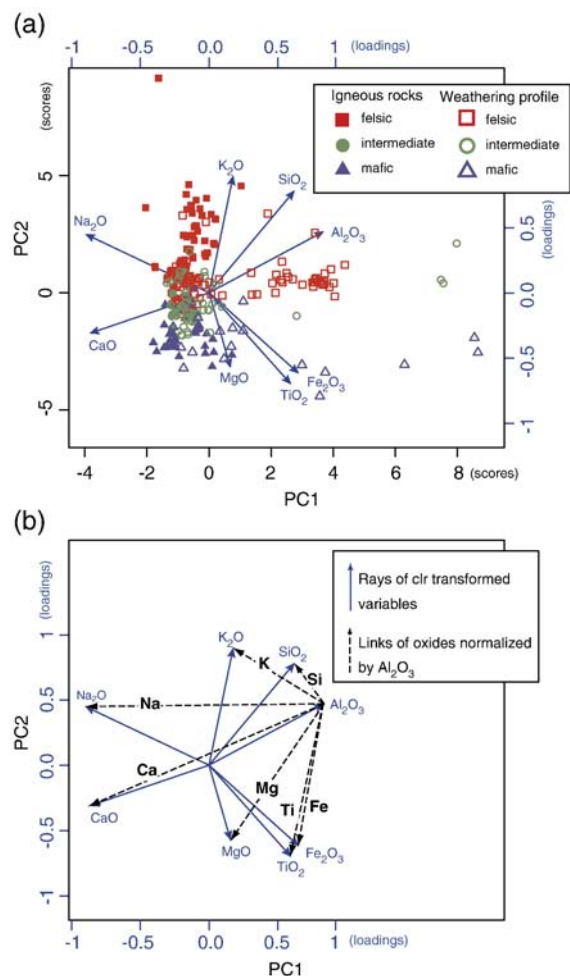


Fig. 1. Relative variation biplot of the clr-transformed data for fresh and weathered rocks compiled from the literature (Table 2). (a) Scores for samples and loadings of oxides. Compositional variations due to weathering correlate well with PC1, whereas those due to the composition of the unweathered parent rocks correlate with PC2. (b) Loadings of oxides and their links. Blue solid arrows indicate loadings of variables. Black dashed arrows indicate links between oxides and Al<sub>2</sub>O<sub>3</sub>, interpreted as the logarithm of oxides normalized by Al<sub>2</sub>O<sub>3</sub>. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In contrast, PC2 is the direction along which the projections of the felsic, intermediate and mafic igneous rocks appear to be best discriminated (Fig. 1a). The links of elements normalized by  $\text{Al}_2\text{O}_3$  show that PC2 positively covaries with **K** and **Si** and negatively covaries with **Ti**, **Mg**, **Ca** and **Fe**. Consequently, the PC2 coordinate appears to separate felsic rocks that are relatively enriched in  $\text{K}_2\text{O}$  and  $\text{SiO}_2$  from mafic rocks that are enriched in  $\text{TiO}_2$ ,  $\text{MgO}$ ,  $\text{CaO}$  and  $\text{Fe}_2\text{O}_3$ . This is consistent with the commonly held views on the nature of geochemical variations across the series from felsic to mafic igneous rocks (e.g., Rollinson, 1993). On the basis of these observations, PC2 is interpreted to reflect chemical variability within igneous rocks.

### 3.2. MFW diagram

The results of the PCA indicate that geochemical variability is governed chiefly by the three latent end members, i.e., fresh mafic source, fresh felsic source and weathered rocks, which can be plotted on a ternary diagram to aid in their interpretation. We therefore further reform PC1 and PC2 coordinate space to introduce a new ternary diagram as a discrimination diagram for chemical weathering. In mapping the PC1 and PC2 by the ilr transformation,  $v_1$  in Eq. (A4) was substituted by  $-\text{PC2}$  and  $v_2$  was substituted by  $-\text{PC1}$ . As the positive and negative signs of the PCA are arbitrary, changing it does not alter the data. Note that the ilr transformation maintains the independence of

coordinates (Egozcue et al., 2003), indicating that PC1 and PC2 will remain geochemically independent on the transformed ternary diagram.

The ilr transformation produces three latent variable vertices, *M*, *F* and *W* representing the mafic source, felsic source and weathered material, respectively. Fig. 2 shows the MFW diagram and a plot of the input data listed in Table 2. Unweathered igneous samples plot near the *M*–*F* join, mafic rocks plot in the upper part of the diagram, felsic rocks plot in the lower left region and intermediate rocks plot in the central area (Fig. 2a). Weathered samples extend toward the *W* vertex and highly weathered samples plot close to the *W* vertex (Fig. 2b). Therefore, the *M* vertex typifies mafic parent rock, the *F* vertex typifies felsic parent rock and the *W* vertex identifies the degree of weathering of the rocks.

The equations used to calculate the *M*, *F* and *W* vertices are listed in Table 4. As is shown in the following results, these equations are not only valid for the current input database but also for most homogenous igneous rocks and weathering profiles when plotting data on a ternary diagram (e.g., Fig. 3).

To validate the interpretation of the *M*, *F* and *W* vertices, additional data that were not used in the development of the model were plotted upon the MFW diagram (Fig. 3). Igneous rocks ranging from felsic to mafic and ultramafic define a compositional linear trend that extends between the *M* and *F* vertices (solid line in Fig. 3a). Most of the variance in the igneous rocks is explained by this compositional linear trend (99.03%).

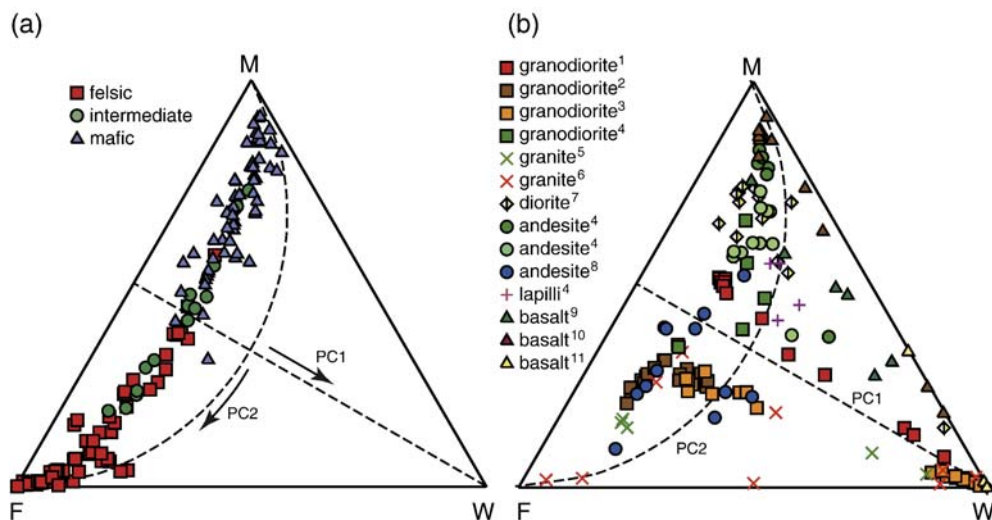


Fig. 2. MFW ternary plot for igneous rocks (a) and weathering profiles (b). Dashed lines are traces of the PC1 and PC2 coordinates on the ternary plot. Superscript figures in the legend correspond to the following references: (1) Nesbitt and Markovics (1997); (2) Girty et al. (2003); (3) White et al. (2001); (4) Lee et al. (2004); (5) Mongelli (1993); (6) Lan et al. (2003); (7) Turner et al. (2003); (8) Ece and Nakagawa (2003); (9) Price et al. (1991); (10) Moon and Jayawardane (2004); (11) Soubrand-Colin et al. (2005).

Table 4  
Formulas for calculating vertices

## STEP 1

$$M = -0.395 \times \ln(\text{SiO}_2) + 0.206 \times \ln(\text{TiO}_2) - 0.316 \times \ln(\text{Al}_2\text{O}_3) + 0.160 \times \ln(\text{Fe}_2\text{O}_3) + 0.246 \times \ln(\text{MgO}) + 0.368 \times \ln(\text{CaO}^*) + 0.073 \times \ln(\text{Na}_2\text{O}) - 0.342 \times \ln(\text{K}_2\text{O}) + 2.266$$

$$F = 0.191 \times \ln(\text{SiO}_2) - 0.397 \times \ln(\text{TiO}_2) + 0.020 \times \ln(\text{Al}_2\text{O}_3) - 0.375 \times \ln(\text{Fe}_2\text{O}_3) - 0.243 \times \ln(\text{MgO}) + 0.079 \times \ln(\text{CaO}^*) + 0.392 \times \ln(\text{Na}_2\text{O}) + 0.333 \times \ln(\text{K}_2\text{O}) - 0.892$$

$$W = 0.203 \times \ln(\text{SiO}_2) + 0.191 \times \ln(\text{TiO}_2) + 0.296 \times \ln(\text{Al}_2\text{O}_3) + 0.215 \times \ln(\text{Fe}_2\text{O}_3) - 0.002 \times \ln(\text{MgO}) - 0.448 \times \ln(\text{CaO}^*) - 0.464 \times \ln(\text{Na}_2\text{O}) + 0.008 \times \ln(\text{K}_2\text{O}) - 1.374$$

## STEP 2

Closure operation<sup>a</sup>:  $C_{100}[\exp(M), \exp(F), \exp(W)]$

In STEP 1, the units for the oxides are in wt.%. Note that eight oxides used in the formulas must be re-closed to 100 wt.% before calculation. If the quantity of carbonate and apatite in the sample is significant, CaO attributable to these should also be corrected (denoted as CaO\*). In STEP 2, take exponents of “M”, “F” and “W” values and re-total these three values to 100.

<sup>a</sup> Closure operation  $C_{100}()$  denotes normalization to composition summing to 100 (see Appendix).

Thus, Fig. 3a supports the interpretation that the *M* and *F* vertices represent the geochemical variability of igneous rocks.

Fig. 3b provides examples of modern weathering profiles developed within basalt (Middle Atlas, Morocco; Dekayir and El-Maataoui, 2002), diorite (northern

Honshu, Japan; Nakagawa et al., 1972) and granite (South Carolina, USA; Rudnick et al., 2004). Note that the fresh basalt sample plots close to the igneous rock trend. The basaltic weathering profile shows a compositional linear trend that extends from the *M* vertex to the *W* vertex (Fig. 3b). Dekayir and El-Maataoui (2002) classified the degree of weathering into four categories on the basis of visual inspection: facies 1, fresh core basalt; facies 2, intermediately weathered samples; facies 3, extensively weathered samples; facies 4, soil. Fresh samples (facies 1) to intermediately weathered samples (facies 2) generally plot near the *M* vertex, whereas extensively weathered samples (facies 3) extend toward the *W* vertex and soil samples (facies 4) plot close to the *W* vertex.

The weathering profile of diorite also extends toward the *W* vertex. The extent of weathering, which was determined by the abundance of vermiculite, kaolinite and gibbsite, increases progressively from the fresh S1 sample to the weathered S6 sample (Nakagawa et al., 1972). Accordingly, increasing *W* values represent an increasing abundance of secondary weathering-related minerals.

The granitic weathering profile also shows a compositional linear trend extending from the *F* vertex to the *W* vertex. In this profile, the bulk density decreases with increasing intensity of weathering (Rudnick et al.,

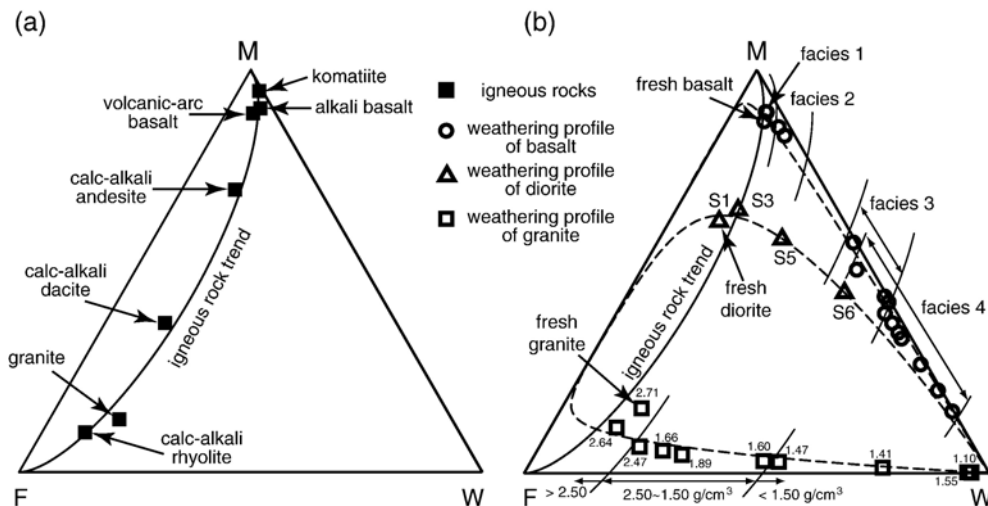


Fig. 3. The MFW plots for representative igneous rocks (a) and weathering profiles (b). The line in (a) represents a compositional linear trend for igneous rocks, which explains 99.03% of the variability induced by the igneous rocks. Data for igneous rocks are from Arndt et al. (1977), Macdonald (1968), Jakes and White (1972) and Ewart (1982) and Phillips et al. (1981). The broken lines in (b) are compositional linear trends for weathering profiles of basalt, diorite and granite (Nakagawa et al., 1972; Dekayir and El-Maataoui, 2002; Rudnick et al., 2004). Samples from the basalt profile are classified into the following facies; fresh (facies 1), intermediately weathered (facies 2), extensively weathered (facies 3) to soil (facies 4). In the diorite profile, the proportions of secondary weathering minerals increase in the following order; S1, S3, S5 to S6. The numbers along the granite weathering profile indicate the bulk densities of the samples, which provide an estimate of the degree of weathering. Note that a decrease in the bulk density generally corresponds with an increase in the *W* value.



2004). A fresh granite sample with a high bulk density ( $2.71 \text{ g/cm}^3$ ) plot close to the igneous rock trend (Fig. 3b), whereas a highly weathered saprolite sample with a low bulk density ( $1.10 \text{ g/cm}^3$ ) plot close to the  $W$  vertex (Fig. 3b). Although there are some exceptions, samples with lower bulk densities generally plot closet to the  $W$  vertex. These observations demonstrate that the linear trends apparent in Fig. 3b represent weathering trends and that the  $W$  values provide a measure of the weathering intensity.

For these weathering profiles, the amount of variance explained by the compositional linear trends (broken lines) were 86.66%, 94.60% and 98.27% for the basalt, diorite and granite profiles, respectively. Despite the differences in the magmatic origins and petrographic characteristics of the unweathered parent rocks, all of the weathering trends are linearly expressed on the MFW diagram and extend toward the  $W$  vertex. This property is a direct benefit of the  $\text{ilr}$  transformation, which preserved the independence of the PC1 and PC2. Accordingly, the variability of unweathered parent rocks is irrelevant in the present scheme and all of the weathering trends are comparable, extending to the  $W$  vertex on the MFW diagram.

## 4. Discussion

### 4.1. Evidence of the weathering of rock-forming minerals on the MFW diagram

Price and Velbel (2003) noted that a useful weathering index should exhibit appropriate chemical trends with progressive weathering. The simplest way to conceptualize the weathering process is to describe the stoichiometry of the decomposition of plagioclase to kaolinite. Table 5 summarizes variations in the WIP, CIA and  $W$  values that accompany the dissolution of plagioclase. Note that the index values for CIA and  $W$  increase with progressive weathering, whereas WIP decreases. The CIA values are entirely consistent with

the dissolution of plagioclase in that anorthite and albite yield identical CIA values that are linearly proportional to the degree of dissolution. In contrast, the WIP and  $W$  indices yield values that do not vary linearly with the dissolution of anorthite and albite. However, the behavior of the  $W$  index is consistent with the weathering processes of plagioclase.

Processes of mineral weathering can also be ascertained graphically from the MFW diagram (Fig. 4). In general, mafic minerals demonstrate high  $M$  values (except for anorthite), felsic minerals record high  $F$  values and phyllosilicates demonstrate high  $W$  values (Fig. 4). This result for silicate minerals is consistent with the above interpretations of the three vertices. Exceptions occur for some FeO-rich primary minerals such as biotite, fayalite, ferrosilite and chlorite (Fig. 4). These minerals plot close to the  $W$  vertex because the present scheme does not discriminate between the FeO and  $\text{Fe}_2\text{O}_3$  components. This pattern suggests that biotite-, fayalite-, ferrosilite- and chlorite-rich rocks would yield high  $W$  values regardless of whether they were fresh or weathered. However, the basalt, dolerite and gabbro reference samples shown in Fig. 2 do not show considerable changes in  $W$  values unless weathered. Thus, the effect of these minerals upon the net  $W$  value is limited except in the rare case where these minerals are the predominant constituent of the rock.

Plotted points for unweathered plagioclases show that an increase in the  $W$  value indicates translation along the  $F$ – $W$  join toward the  $W$  vertex, which passes through the compositional region of kaolinite (Fig. 4b). Therefore, the observed increase in the  $W$  value corresponds exactly to the path of mineralogical decomposition from plagioclases to kaolinite. Similarly, the transformation of muscovite to illite during weathering is expressed as an increase in the  $W$  value (Fig. 4b) and translation along the  $M$ – $W$  join corresponds exactly to the alteration of mafic minerals to saponite or nontronite (Fig. 4). In brief, an increase in the  $W$  value can be explained by the progressive loss of primary minerals and corresponding increase in the abundance of secondary phyllosilicates.

To summarize, the main limitation of the  $W$  index relative to the CIA is that plagioclase dissolution cannot be directly quantified. However, the CIA does not account for the path of mineralogical dissolution of unweathered parent rocks with mixed mineralogy. The  $W$  and WIP indices are superior in this regard because they are not designed to reflect the decomposition of a single mineral, rather they reflect the collective decomposition of multiple mineral phases.

Table 5  
Plagioclase dissolution and index values

| Degree of dissolution (%) | Anorthite |       |      | Albite |       |      |
|---------------------------|-----------|-------|------|--------|-------|------|
|                           | WIP       | CIA   | W    | WIP    | CIA   | W    |
| 0                         | 51.3      | 50.0  | 8.7  | 108.9  | 50.0  | 0.9  |
| 20                        | 42.8      | 55.6  | 9.7  | 98.5   | 55.6  | 1.0  |
| 40                        | 33.5      | 62.5  | 11.1 | 84.9   | 62.5  | 1.3  |
| 60                        | 23.4      | 71.4  | 13.3 | 66.6   | 71.4  | 1.5  |
| 80                        | 12.2      | 83.3  | 18.1 | 40.4   | 83.3  | 2.4  |
| 100                       | 0.0       | 100.0 | 89.6 | 0.0    | 100.0 | 89.6 |

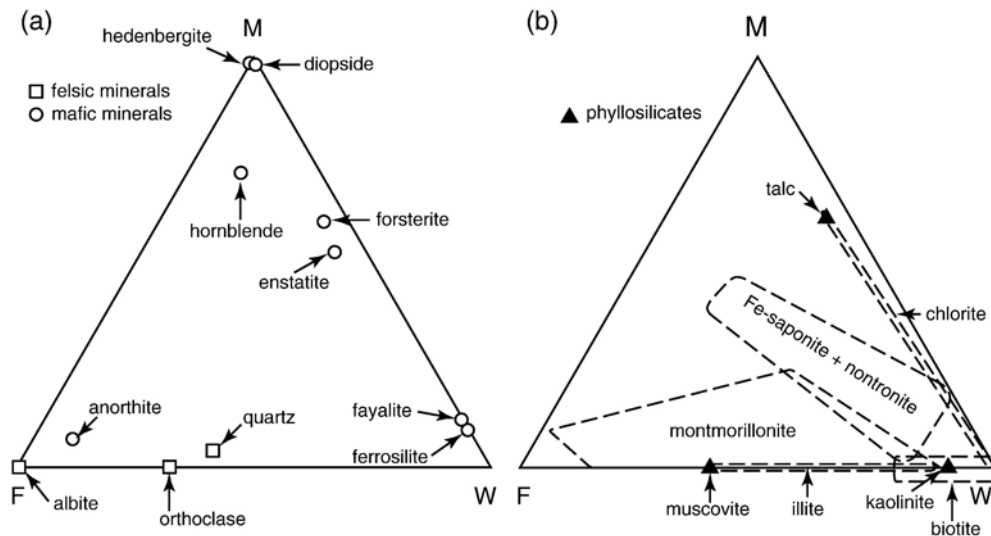


Fig. 4. MFW plots of typical rock-forming minerals (a) and phyllosilicates (b). Felsic minerals define a cluster near the *F* vertex, whereas mafic minerals generally cluster close to the *M* vertex and phyllosilicates cluster around the *W* vertex.

#### 4.2. Evaluation of weathering intensity using the *W* index

The above argument demonstrates that the *W* value exhibits appropriate trends with mineral weathering processes. In this section, we further assess the predictive performance of the *W* value using saprolite profiles. Table 6 lists representative *W* values for recent weathering profiles. Central values (in this case, the geometric mean) increase in sequential order from fresh

rock, partly to totally altered rock, saprolite, soil and bauxite. The differences in *W* values for these lithologies are statistically significant because the 95% confidence intervals for the geometric means do not overlap (Table 6). Therefore, in this case the predictive performance of the *W* index is sufficient to enable the discrimination of various weathered materials.

We integrated the *W* index into the framework described by Duzgoren-Aydin et al. (2002) to compare the predictive performance of the *W* index relative to that of other indices. Duzgoren-Aydin et al. (2002) investigated the predictive performance of 27 geochemical weathering indices using a profile of weathered felsic pyroclastic rock developed under humid, relatively well-drained conditions in Hong Kong. These analyzed samples range in weathering grade from fresh (grade 1) to moderately weathered (grade 3), moderately to highly weathered (grade 3–4), highly to completely weathered (grade 4–5) and completely weathered samples (grade 5). Completely weathered samples of grade 5 are further subdivided based on their sesquioxide content. These distinctions in weathering grade were made on the basis of visual assessment in the field and petrographic observations (Duzgoren-Aydin et al., 2002).

The relationships between CIA, WIP and *W* values and weathering grade are shown in Fig. 5. The *W* index shows an abrupt increase from the fresh zone (grade 1) to the moderately weathered zones (grades 3, 3–4); thereafter, the index shows a continued increase as

Table 6  
*W* values for saprolite and bauxite profiles

|                                                | Geometric mean | Minimum | Maximum | 95% confidence interval |             |
|------------------------------------------------|----------------|---------|---------|-------------------------|-------------|
|                                                |                |         |         | Lower limit             | Upper limit |
| Fresh igneous rocks ( <i>N</i> =15)            | 10.68          | 5.99    | 18.51   | 10.48                   | 10.88       |
| Partly to totally altered rock ( <i>N</i> =34) | 33.14          | 12.78   | 98.30   | 31.80                   | 34.56       |
| Saprolite ( <i>N</i> =29)                      | 94.10          | 60.99   | 99.59   | 93.59                   | 94.58       |
| Soil ( <i>N</i> =10)                           | 94.92          | 86.36   | 98.89   | 94.69                   | 95.16       |
| Bauxite ( <i>N</i> =20)                        | 97.42          | 79.28   | 99.01   | 96.81                   | 97.42       |

Data are adopted from Hamdan and Burnham (1996), Hieronymus et al. (2001), White et al. (2001) and Girty et al. (2003).

95% confidence intervals for geometric means were calculated by the bootstrap method.

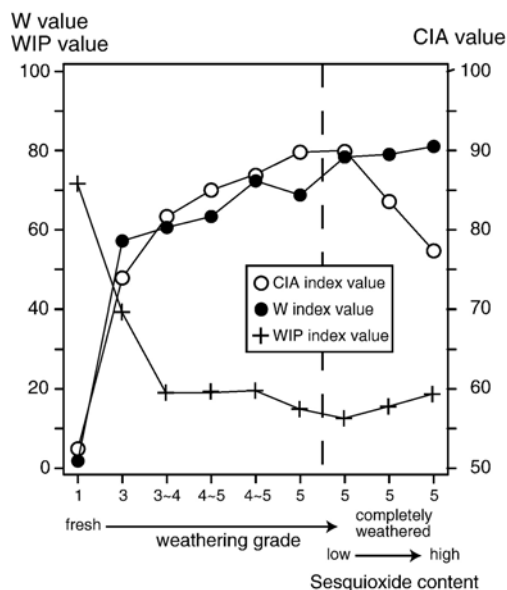


Fig. 5. Variation in the CIA,  $W$  and WIP values with weathering grade and abundance of sesquioxide contents for the weathered felsic pyroclastic rocks profile described by Duzgoren-Aydin et al. (2002). Completely weathered samples of grade 5 are subdivided depending on sesquioxide content. The vertical dashed line separates regions, in which the weathering indices are plotted against the weathering grade and the region intended to demonstrate the effects of sesquioxide content. Note that strong correlation between  $W$  values and weathering grade.

weathering grade increases to the intensively weathered zone (grade 5). According to Duzgoren-Aydin et al. (2002), weathering indices that are not sufficiently sensitive to capture the variations that occur during weathering display zigzag patterns when plotted against weathering intensity. The absence of such patterns for the plotted  $W$  values suggests that the  $W$  index provides a reliable measure of the degree of weathering.

An important feature of the  $W$  index compared to the WIP and CIA indices is that a good correlation between the index values and weathering grades is sustained in the completely weathered sesquioxide-rich zone (Fig. 5). Note that in this zone the CIA values incorrectly display an arched pattern with an abrupt decrease. The sesquioxide-rich sample (grade 5) is marked by a CIA value comparable with that of the moderately weathered sample (grade 3). The WIP index also displays a weak inverse trend within the sesquioxide-rich zone. Most of the other weathering indices investigated by Duzgoren-Aydin et al. (2002) displayed similar erroneous results.

The primary difference between the  $W$  index and conventional weathering indices is that the  $W$  index is

based on the behaviors of multiple major oxides during weathering. This includes  $\text{Al}_2\text{O}_3$  and  $\text{Fe}_2\text{O}_3$ , which are important in characterizing sesquioxide-rich zones. This accounts for the reason why the  $W$  index returns the most robust result compared to the WIP and CIA indices in the sesquioxide-rich zones, at least, in the example shown.

#### 4.3. Application of the MFW diagram to different rock types

Fedo et al. (1995) and Price and Velbel (2003) stated that a useful weathering index should be applicable to a wide range of rock types and should yield a unique value for the unweathered parent material. The applicability of the MFW diagram to weathering profiles of granite, diorite and basalt has already been demonstrated in Fig. 3b. In Fig. 6, additional weathering profiles are verified. In this figure, weathering profiles for felsic pyroclastics, charnockite and amphibolite are depicted on A-CN-K and MFW diagrams.

The weathering trends for charnockite and amphibolite on the A-CN-K diagram are nearly identical (Fig. 6a). The felsic pyroclastics are the only immediately effective lithology in terms of depicting a distinct weathering trend on the A-CN-K diagram. This is because the A-CN-K diagram considers only the  $\text{Al}_2\text{O}_3$ ,  $\text{CaO}$ ,  $\text{Na}_2\text{O}$  and  $\text{K}_2\text{O}$  sub-composition. If other oxides are considered, these three lithologies will show clear differences in bulk major oxide compositions. In particular, fresh charnockite shows low  $\text{FeO}$  and  $\text{MgO}$  contents ( $\text{FeO}=7.9$  wt.% and  $\text{MgO}=1.3$  wt.%; Sharma and Rajamani, 2001) relative to fresh amphibolite ( $\text{FeO}=9.9$  wt.% and  $\text{MgO}=15.4$  wt.%; Sharma and Rajamani, 2000).

On the A-CN-K diagram, the fresh felsic pyroclastic sample plot close to the plagioclase–potassic feldspar tieline (Pl–Ksp tieline), whereas fresh charnockite and amphibolite samples plot considerably below this line (Fig. 6a). Consequently, the most highly weathered amphibolite sample and unweathered pyroclastic sample yield similar CIA values of about 50. This particular example demonstrates that the CIA needs to be qualified for individual rock types, thus limiting comparisons of absolute CIA values among different parent rocks.

In contrast, the MFW scheme accounts for most major oxides, and weathering profiles for felsic pyroclastics, charnockite and amphibolite are clearly distinguished (Fig. 6b). Despite the fact that the three parent materials have contrasting mineralogy and chemistry, the low  $W$  values for the parent materials

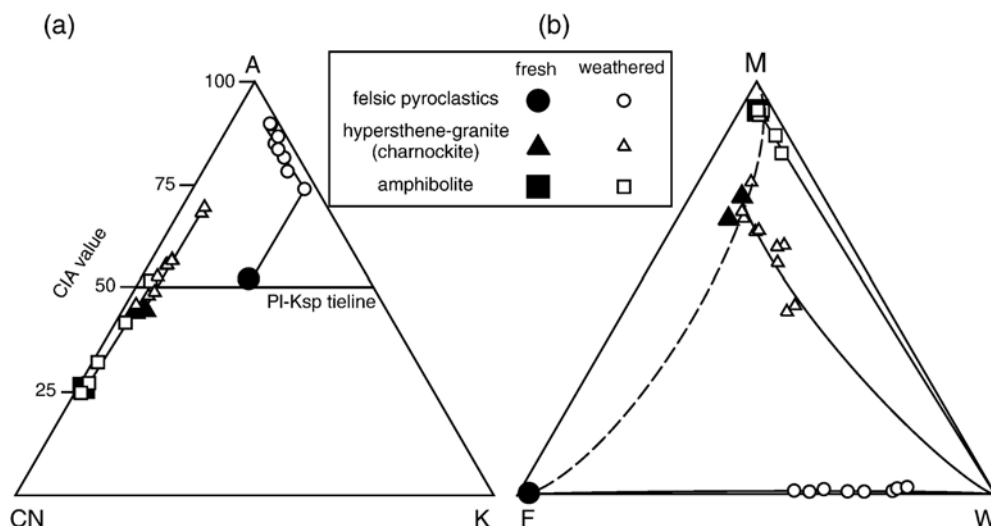


Fig. 6. Comparison of weathering profiles developed within felsic pyroclastics, charnockite and amphibolite on A-CN-K diagram (a) and MFW diagram (b). The vertices of the A-CN-K diagram represent molecular proportions of Al<sub>2</sub>O<sub>3</sub>, CaO + Na<sub>2</sub>O and K<sub>2</sub>O, respectively (Nesbitt and Young, 1984). Data for pyroclastic rocks reported by Duzgoren-Aydin et al. (2002) are the same samples as those used in Fig. 5. The weathering profiles of charnockite and amphibolite are adapted from those described in Sharma and Rajamani (2000, 2001), which developed under semiarid conditions in India. The MFW diagram differentiates the weathering trends of charnockite and amphibolite, whereas the A-CN-K diagram does not. Note also that the unweathered protoliths for the three profiles yield similar *W* values and plot near the igneous trend on the MFW diagram.

are remarkably similar, plotting on the line of the igneous rock trend (Fig. 6b). This suggests that the MFW diagram is applicable to a wide range of weathering profiles and that unweathered parent rocks yield a unique *W* value ranging from 0 to 15, regardless of parent rock composition. Thus, the MFW scheme appears to satisfy Price and Velbel's (2003) list of desired attributes for a useful weathering index.

The data presented in Fig. 6b reveal that the *W* values are not always systematic with respect to the weathering profiles, especially for semiarid conditions. There exists significant overlap between fresh charnockite and amphibolite samples and their weathered equivalents. This may well lead to problems in using the *W* index to characterize the weathering grade of a single weathering profile. However, such overlap-related problems are not specific to the MFW diagram, but they also occur in the A-CN-K diagram (Fig. 6a).

#### 4.4. Implications for provenance studies

One potential application of the MFW diagram is provenance analyses of clastic sediments, as with the A-CN-K diagram. The mineralogy and geochemistry of sediments commonly reflect the compositions of weathering profiles of the source rock, rather than that of the source rock itself (e.g., Nesbitt et al., 1996). Accordingly, a direct interpretation of provenance

based on sediment composition may not adequately reflect the composition of the source rock (e.g., Johnsson, 1993).

In this context, Fedo et al. (1995), Nesbitt and Young (1996) and Nesbitt et al. (1996) described an innovative approach to provenance analysis that implicitly considers geochemical modifications associated with weathering. In a similar manner, the MFW diagram can overcome the weathering effect by predicting the original composition of the source rock. Samples of sediment corrected from a weathering profile may delineate a compositional trend on the MFW diagram. In this case, the composition of the unweathered source rock can be reconstructed simply by tracing the trend backward to the unweathered domain (von Eynatten, 2004). Therefore, the utility of the MFW diagram is not restricted to the assessment of weathering profiles, but it may also be helpful in facilitating provenance analysis.

#### 4.5. Limitations of the MFW scheme

Two important variables that are not evaluated in the MFW scheme are the role of volatile components and the redox behavior of iron. Weathering products are commonly hydrous, leading to significant variations in H<sub>2</sub>O content during the course of weathering. Incorporation of the LOI component into the analysis is



therefore expected to increase the sensitivity and the accuracy of the  $W$  index. Another important component that the MFW scheme disregards is the distinction between ferric and ferrous iron. It is known that FeO is released through hydrolysis and oxidation processes during the weathering of silicate minerals, and thus, Fe<sub>2</sub>O<sub>3</sub> content relative to FeO is clearly linked to the degree of weathering (e.g., Railsback, 2003). von Eynatten et al. (2003) reported that the results of PCA are subject to marked change if these components are neglected. However, in the present study, the  $W$  value has been successfully applied to some weathering profiles and clearly provides insight into the processes that occur during weathering. Therefore, our interpretation that the  $W$  value measures the degree of weathering remains valid.

A further disadvantage of the MFW diagram is that it is impossible to speculate on the stoichiometric significance of the  $W$  index. The MFW scheme involves rather complicated mathematics and the final equation is complex. In contrast, the stoichiometric significance of the CIA index is clear, showing a linear response to the molecular decomposition of feldspars (Table 5). However, conventional indices may yield anomalous values depending on the parental rock type and the physico-chemical conditions of the weathering environment (e.g., Fig. 5). It is, therefore, necessary to combine the results of numerous indices based on different criteria to obtain a representative solution. The  $W$  index can be applied in a variety of settings because it is determined from a dataset that includes most of the common igneous rock types from a variety of weathering environments.

The  $W$  index is based on data from homogenous igneous parent rocks and their weathering profiles. Price and Velbel (2003) found that indices established from such profiles commonly underperform in terms of characterizing weathering profiles developed within heterogeneous parent rocks (e.g., metamorphic rocks). Indeed, chlorite, which is a primary metamorphic mineral, has a high  $W$  value (Fig. 4b) suggesting that the MFW diagram may not be applicable to weathering profiles developed within greenschist facies metamorphic rocks. The applicability of the  $W$  index to weathering profiles developed within metamorphic rocks needs to be examined in future studies.

## Appendix A. Concepts of compositional data analysis

### A.1. Simplex; a sample space of compositional data

In statistical modeling, it is important to recognize a correct sample space of data in a given subject. A sample space is a collectively exhaustive listing of all possible outcomes of an experiment or event. Note that the Euclidean real sample space ( $\mathbb{R}^D$ ) is not an appropriate sample space for compositional data: the appropriate space is a unit

## 5. Conclusion

The present study proposes a new chemical weathering index, the  $W$  index, and associated MFW ternary diagram for the visual presentation of weathering trends. The  $W$  index is extracted using statistical empirical laws, and possesses a number of unique properties that are not found in conventional indices. We verified that an increase in the  $W$  value is concordant with the weathering processes of multiple-mineral phases. The  $W$  index also returns robust results even for highly weathered sesquioxide-rich materials. Most importantly, we have showed that the  $W$  index is widely applicable to most igneous rock types. The MFW diagram is expected to facilitate provenance analysis of sedimentary rocks by predicting their weathering trends and enabling an evaluation of the composition of unweathered source rocks.

At the expense of these benefits, however, the stoichiometric significance of the  $W$  index is complex and no simple chemical reaction or single mineral alteration process is responsible for variation in the  $W$  index. An explicit weathering index that is capable of explaining both the dissolution of a single mineral (e.g., the CIA index) and the bulk geochemical response during weathering (e.g., the  $W$  index) does not exist. It is important to stress that the objective of the  $W$  index lies in generating a picture of the bulk geochemical response to the weathering process regardless of parent rock types rather than the dissolution process of a single mineral.

## Acknowledgements

We are indebted to Dr. Timothy J. Fagan for the constructive discussions and for reviewing an earlier version of this manuscript. Comments and helpful suggestions made by the editor Dr. Roberta L. Rudnick and the reviewers Dr. Jason R. Price and anonymous reviewer greatly enhanced the quality of this manuscript. We would like to thank Dr. Richard Reymont and Dr. Yukiyasu Saka for discussions about the compositional data analysis and many suggestions from which this paper greatly benefited.

simplex sample space (Aitchison, 1982). The simplex sample space for  $D$ -part compositional data  $\mathbf{x}=(x_1, \dots, x_D)$  is defined as

$$\mathbb{S}^D = \left\{ \mathbf{x} = [x_1, \dots, x_D] \mid x_i > 0, i = 1, \dots, D, \sum_{i=1}^D x_i = 100 \right\} \quad (\text{A1})$$

The definition of the simplex (A1) indicates that compositional data actually lie within a  $(D-1)$ -dimensional space and that the degree of freedom is one less than the number of variables. For example, consider a composition with two variables ( $D=2$ ). If the value of one variable is determined, the value of the other is determined automatically by the subtraction of the first variable from 100. Although we have two variables in this case, the information that we possess is really just one-dimensional. It follows that variables of compositional data are not free to vary independently. Since the independence of variables is not valid, statistical concepts such as regression and correlation cannot be interpreted equivalently as real number data associated with the Euclidean sample space (e.g., Chayes, 1960).

## A.2. Logratio analysis

To overcome this problem, Aitchison (1982) and Egozcue et al. (2003) proposed data transformation techniques that can be used to map the compositional data from the simplex to the Euclidean real sample space. Among these, we utilized the centered logratio transformation (clr) and the isometric logratio transformation (ilr). For a  $D$ -part compositional data  $\mathbf{x}=(x_1, \dots, x_D)$ , clr and ilr are defined as

$$\text{clr}(\mathbf{x}) = \left[ \ln \frac{x_1}{(\prod_{i=1}^D x_i)^{\frac{1}{D}}}, \ln \frac{x_2}{(\prod_{i=1}^D x_i)^{\frac{1}{D}}}, \dots, \ln \frac{x_D}{(\prod_{i=1}^D x_i)^{\frac{1}{D}}} \right] = \mathbf{z}(z_1, \dots, z_D) \quad (\text{A2})$$

$$\text{ilr}(\mathbf{x}) = \left[ \sqrt{\frac{1}{2}} \ln \frac{x_1}{x_2}, \sqrt{\frac{2}{3}} \ln \frac{\sqrt{x_1 x_2}}{x_3}, \dots, \sqrt{\frac{D-1}{D}} \ln \frac{(\prod_{i=1}^{D-1} x_i)^{\frac{1}{D-1}}}{x_D} \right] = \mathbf{v}(v_1, \dots, v_{D-1}) \quad (\text{A3})$$

Eqs. (A2) and (A3) convert a compositional variable that must be strictly positive and finite (0~100%) into a variable that can take any real values ( $-\infty \sim \infty$ ). Furthermore, it is known that logratio-transformed data tend to follow a multivariate normal distribution (e.g., Aitchison and Shen, 1980). Thus, statistical techniques that require multivariate normality can be employed following logratio transformation. In fact, the application of such logratio techniques to geological data is increasing in popularity (e.g., Zhou et al., 1991; Cardenas et al., 1996; Kucera and Malmgren, 1998; Reyment and Hirano, 1999; Sircombe, 2000; Walvoort and de Gruijter, 2001).

An important aspect of logratio transformations is that they offer a way to convert compositional data to real number data, and vice versa. For example, the inverse of ilr for a two-part real vector,  $\mathbf{v}=(v_1, v_2)$ , is achieved by

$$\text{ilr}^{-1}(\mathbf{v}) = C_{100} \left[ \exp \left( \frac{v_2}{\sqrt{6}} + \frac{v_1}{\sqrt{2}} \right), \exp \left( \frac{v_2}{\sqrt{6}} - \frac{v_1}{\sqrt{2}} \right), \exp \left( \frac{-2v_2}{\sqrt{6}} \right) \right] \quad (\text{A4})$$

where  $C_{100}()$  denotes the closure operator, which describes the normalization of components to percentages;

$$C_{100}(\mathbf{v}) = \left( \frac{v_1}{\sum v_i}, \dots, \frac{v_D}{\sum v_i} \right) \times 100 \quad (\text{A5})$$

In the present paper, we optimized this bijective property of the clr and ilr transformation to undertake conventional statistical analyses of compositional data and express the results in terms of compositions. The exact sequence of

operations involved in the present analysis is described in Table A1. First, to ensure that the input data are suitable for PCA, compositional data were transformed to ordinary real data using clr (Step 1 in Table A1). In this step, we used the clr transformation because it returns a unique real vector that maintains the same number of components as those in the original compositional data. To compute clr in Eq. (A2), we first calculated the geometric mean. Consider the three-part composition  $\mathbf{x}=(\text{SiO}_2, \text{Al}_2\text{O}_3, \text{Fe}_2\text{O}_3)=(55.5, 24.2, 20.3)$ , the geometric mean for this composition is

$$g(\mathbf{x}) = \sqrt[3]{55.5 \times 24.2 \times 20.3} = 30.1$$

clr( $\mathbf{x}$ ) is obtained by taking the logarithm of components divided by the geometric mean;

$$\text{clr}(\mathbf{x}) = \left[ \ln \frac{55.5}{30.1}, \ln \frac{24.2}{30.1}, \ln \frac{20.3}{30.1} \right] = (0.612, -0.218, -0.394)$$

In the second step, a standard PCA was applied to the clr-transformed data (Step 2 in Table A1). The third step reformulates these orthogonal components into compositions (ternary diagram). The ilr transformation is a specialized transformation for compositional data that selects orthonormal bases in the Euclidean sample space. Therefore, the

Table A1. Scheme illustrating the sequence of operation followed by this study

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| STEP 1   | Operation: clr transformation<br>Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sample space: $\mathbb{S}^8 \rightarrow \mathbb{R}^8$ |
|          | $\text{clr}(\mathbf{x}) = \ln \left( \frac{x_i}{g(\mathbf{x})} \right) \quad i = 1, \dots, D$ $= [\text{clr}(\text{SiO}_2), \text{clr}(\text{TiO}_2), \text{clr}(\text{Al}_2\text{O}_3), \text{clr}(\text{Fe}_2\text{O}_3), \text{clr}(\text{MgO}), \text{clr}(\text{CaO}), \text{clr}(\text{Na}_2\text{O}), \text{clr}(\text{K}_2\text{O})]$ $\text{note: } g(\mathbf{x}) = [(\text{SiO}_2) \cdot (\text{TiO}_2) \cdot (\text{Al}_2\text{O}_3) \cdot (\text{Fe}_2\text{O}_3) \cdot (\text{MgO}) \cdot (\text{CaO}) \cdot (\text{Na}_2\text{O}) \cdot (\text{K}_2\text{O})]^{1/8}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                       |
| STEP 2   | Operation: PCA<br>Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sample space: $\mathbb{R}^8 \rightarrow \mathbb{R}^2$ |
|          | $\text{PC1} = 0.334 [\text{clr}(\text{SiO}_2) - \overline{\text{clr}(\text{SiO}_2)}] + 0.319 [\text{clr}(\text{TiO}_2) - \overline{\text{clr}(\text{TiO}_2)}] + 0.448 [\text{clr}(\text{Al}_2\text{O}_3) - \overline{\text{clr}(\text{Al}_2\text{O}_3)}] + 0.349 [\text{clr}(\text{Fe}_2\text{O}_3) - \overline{\text{clr}(\text{Fe}_2\text{O}_3)}]$ $+ 0.082 [\text{clr}(\text{MgO}) - \overline{\text{clr}(\text{MgO})}] - 0.463 [\text{clr}(\text{CaO}) - \overline{\text{clr}(\text{CaO})}] + 0.484 [\text{clr}(\text{Na}_2\text{O}) - \overline{\text{clr}(\text{Na}_2\text{O})}] + 0.095 [\text{clr}(\text{K}_2\text{O}) - \overline{\text{clr}(\text{K}_2\text{O})}]$ $\text{PC2} = 0.445 [\text{clr}(\text{SiO}_2) - \overline{\text{clr}(\text{SiO}_2)}] + 0.395 [\text{clr}(\text{TiO}_2) - \overline{\text{clr}(\text{TiO}_2)}] + 0.268 [\text{clr}(\text{Al}_2\text{O}_3) - \overline{\text{clr}(\text{Al}_2\text{O}_3)}] - 0.348 [\text{clr}(\text{Fe}_2\text{O}_3) - \overline{\text{clr}(\text{Fe}_2\text{O}_3)}]$ $+ 0.315 [\text{clr}(\text{MgO}) - \overline{\text{clr}(\text{MgO})}] - 0.174 [\text{clr}(\text{CaO}) - \overline{\text{clr}(\text{CaO})}] + 0.256 [\text{clr}(\text{Na}_2\text{O}) - \overline{\text{clr}(\text{Na}_2\text{O})}] + 0.508 [\text{clr}(\text{K}_2\text{O}) - \overline{\text{clr}(\text{K}_2\text{O})}]$ |                                                       |
| STEP 3-1 | Operation: inverse of ilr transformation<br>Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                       |
|          | $V_1 = \exp \left( \frac{-1}{\sqrt{6}} \text{PC1} + \frac{-1}{\sqrt{2}} \text{PC2} \right), \quad V_2 = \exp \left( \frac{-1}{\sqrt{6}} \text{PC1} - \frac{-1}{\sqrt{2}} \text{PC2} \right), \quad V_3 = \exp \left( \frac{2}{\sqrt{6}} \text{PC1} \right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                       |
| STEP 3-2 | Operation: closure operation<br>Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sample space: $\mathbb{R}^3 \rightarrow \mathbb{S}^2$ |
|          | $C_{100}(V_1, V_2, V_3) = \left\{ \frac{V_1}{\sum V_i}, \frac{V_2}{\sum V_i}, \frac{V_3}{\sum V_i} \right\} = (\text{M}, \text{F}, \text{W})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                       |

inverse transformation of *ilr* is the best option for converting PC1 and PC2, which can be considered as orthonormal bases in the Euclidean space. After the inverse mapping of PC1 and PC2 by *ilr*, three latent compositional variables were produced; *M*, *F* and *W*. These variables represent mafic and felsic source rocks and the degree of weathering of these rocks, respectively. Note that we used the orthonormal bases that are equal to PC1 and PC2 but are opposite in sign (Table A1). Ultimately, by following the scheme described in Table A1, the original eight-part oxide compositional data were simplified to three-part latent compositional data that best captured the geochemical variability associated with the lithologies of the unweathered parent rocks and the degree of weathering.

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